

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **Python** services for classification and content quality checks, integrating them into the existing authoring workflows and reducing manual review time by **30%**.
- Developed **REST APIs** in **Java Spring Boot** to power AI-driven documentation experiences, enabling reliable consumption of DITA XML content by AI systems.
- Built **data pipelines** using **Python** and **SQL** to transform and enrich **DITA XML** content, ensuring it was structured and semantically sound for AI indexing and reasoning.
- Collaborated with documentation leaders and platform engineers to define 'AI-ready documentation' standards, focusing on content structure, semantics, and metadata.
- Engineered **Python** utilities for link and structure validation within the documentation toolchain, identifying and flagging over **150** potential structural inconsistencies per release cycle.
- Containerized documentation processing services using **Docker** and deployed them to an **AWS EC2** environment, ensuring scalable and consistent execution of AI-assisted tasks.
- Integrated AI-assisted tooling into **Oxygen XML Editor** and **Git-based workflows**, providing writers with real-time feedback on content quality and structure.

SKILLS

AI/ML/LLMs – LLMs, RAG pipelines, LLM APIs, AI-assisted workflows, Generative AI, Natural Language Processing

Languages – Python, Java, SQL, Bash

Backend/APIs – Spring Boot, REST APIs, FastAPI, Microservices

Data/Content – DITA XML, Structured Content, XML, Data Pipelines, ETL, Content Quality Checks, Metadata

Cloud/DevOps – AWS, EC2, S3, Lambda, CloudWatch, Docker, Git, Jenkins, Azure DevOps Pipelines

Tools – Oxygen XML Editor, Confluence, Jira

PROJECTS

AI-Powered Content Augmentation Pipeline

- Developed a **Python** pipeline to automatically tag and classify documentation sections using an LLM integration, improving search relevance by **20%**.
- Implemented **RAG** principles to enable AI-driven Q&A; capabilities over the DITA XML knowledge base, reducing average query resolution time for internal teams.
- Integrated the pipeline into a **CI/CD** workflow using **Jenkins**, ensuring automated processing of new documentation content.

DITA XML Content Structuring for AI Consumption

- Engineered **Python** scripts to transform and validate DITA XML content, applying semantic metadata and ensuring compliance with a strict schema for AI consumption.
- Built utilities to analyze content structure, identifying and suggesting improvements for chunking and indexing for large language models, covering approximately **8,000** topic files.
- Collaborated with technical writers to gather requirements and iterate on the structuring process, ensuring the output was usable for downstream AI agents.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)